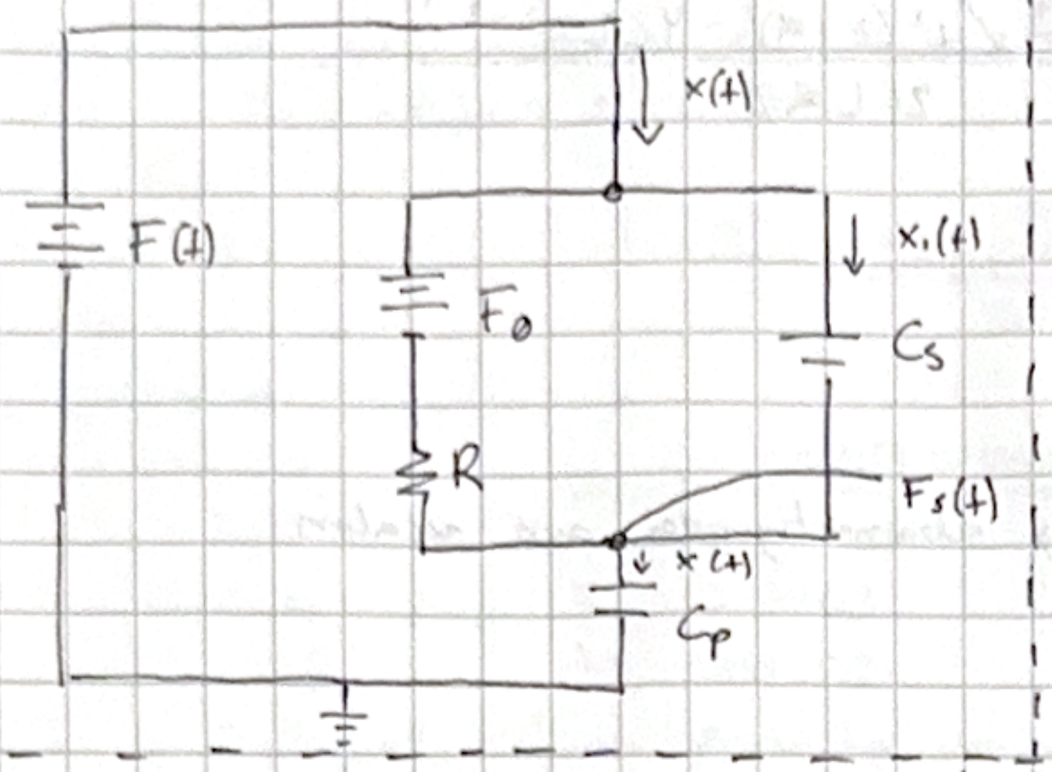
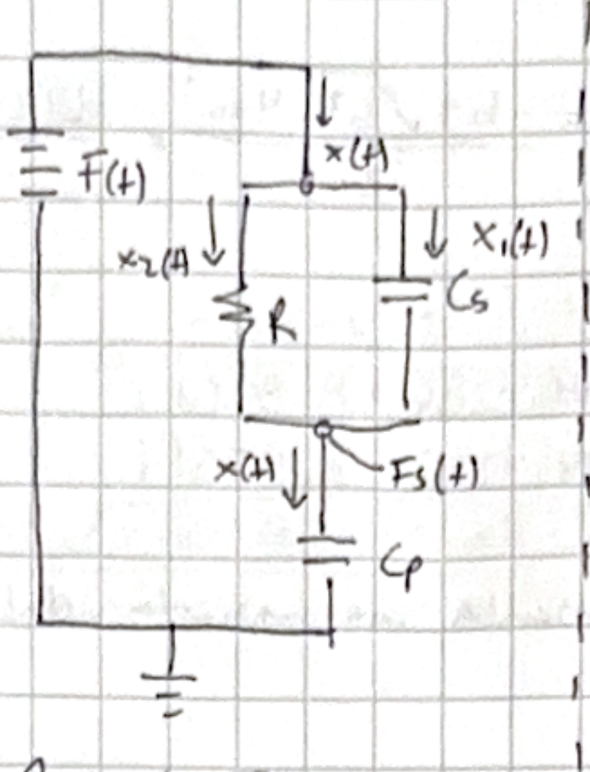
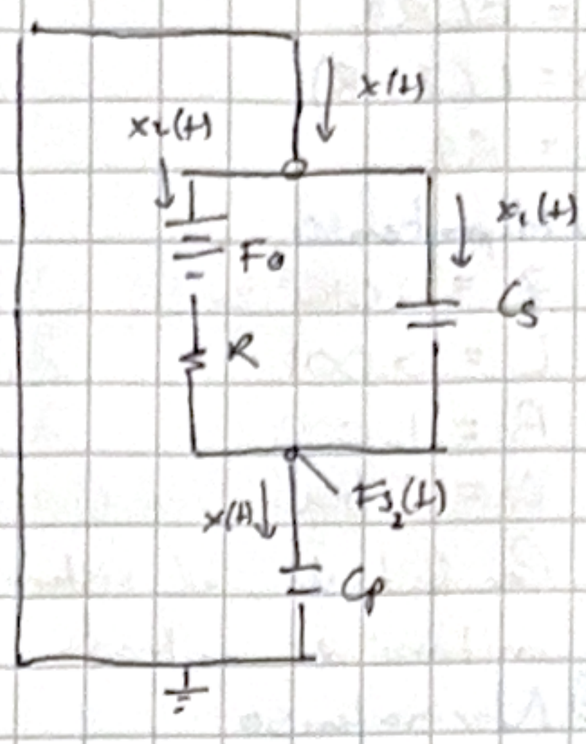
5.5 Práctica: Modulado y control del sistema musculoesquelético

$X(t) = X_1(t) + X_2(t)$

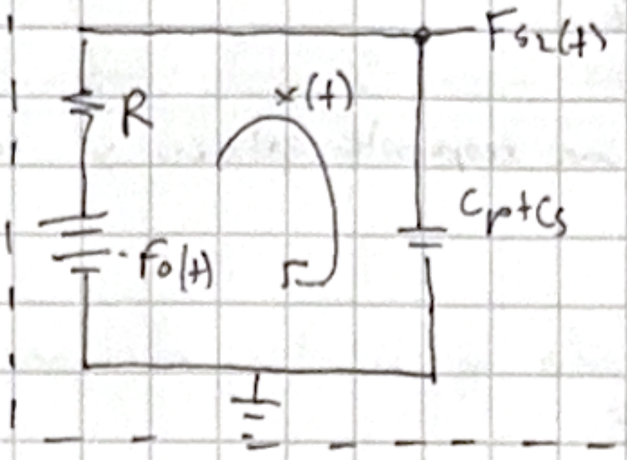




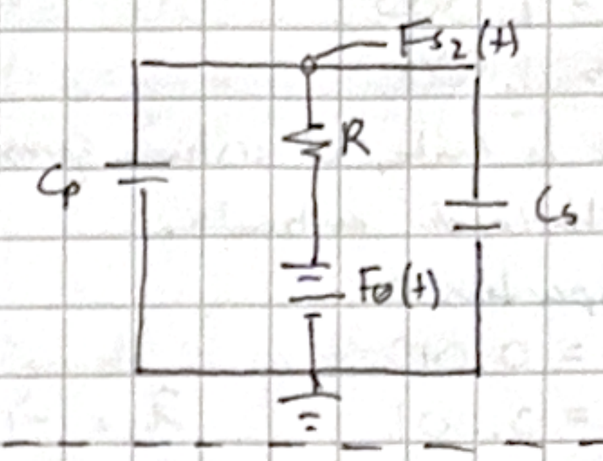
Apagando F_0



Apagando fuente principal



⇐



Análisis matemático

Debido a que el sistema tiene dos fuentes de voltaje se debe aplicar el principio de superposición

Circuito 1: Apagando $F_0(t)$

$$x(t) = x_1(t) + x_2(t)$$

$$x(t) = C_p \frac{dF_{s_1}(t)}{dt}$$

$$x_1(t) = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - F_{s_1}(t)}{R}$$

$$C_p \frac{dF_{s_1}(t)}{dt} = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt} + \frac{F(t) - F_{s_1}(s)}{R}$$

$$C_p s F_{s_1}(s) = C_s s F(s) - C_s s F_{s_1}(s) + \frac{F(s)}{R} - \frac{F_{s_1}(s)}{R}$$

$$C_p s F_{s_1}(s) + C_s s F_1(s) + \frac{F_{s_1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s_1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s_1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s_1}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_p + C_s)s + 1}{R}} F(s)$$

$$F_{s_1}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

Circuito 2: Apagando F(t)

Recordando que $F_0(t) = \alpha F(t)$

$$-\alpha F(t) = R x(t) + \frac{1}{C_p + C_s} \int x(t)\,dt$$

$$F_{s_2}(t) = \frac{1}{C_p + C_s} \int x(t)\,dt$$

$$-\alpha F(s) = R x(s) + \frac{x(s)}{(C_p + C_s)s}$$

$$\bar{F}_{s_2}(s) = \frac{x(s)}{(C_p + C_s)s}$$

$$\bar{F}(s) = -\frac{1}{\alpha}\left(R + \frac{1}{(C_p + C_s)s}\right) x(s) = -\frac{1}{\alpha}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}\,x(s)$$

$$\frac{F_{s_2}(s)}{F(s)} = \frac{\frac{x(s)}{(C_p + C_s)s}}{-\frac{1}{\alpha}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}\,x(s)} = -\frac{\alpha}{R(C_p + C_s)s + 1}$$

$$\bar{F}_{s_2}(s) = -\frac{\alpha F(t)}{R(C_p + C_s)s + 1}$$

Función de transferencia

$$F_s(s) = F_{s_1}(s) + \bar{F}_{s_2}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$\bar{F}_s(s) = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1} F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{RC_s s + 1 - \alpha}{(RC_p + RC_s)s + 1}$$

Error en estado estacionario

$$e(s) = \lim_{s \to 0} F(s)\left(1 - \frac{F_s(s)}{F(s)}\right) = \lim_{s \to 0} \frac{1}{s}\left(1 - \frac{RC_s s + 1 - \alpha}{(RC_p + RC_s)s + 1}\right. = \alpha$$

Estabilidad en lazo abierto

$$(RC_p + RC_s)s + 1 = 0$$

$$\lambda = -\frac{1}{RC_p + RC_s}$$

El sistema es estable.